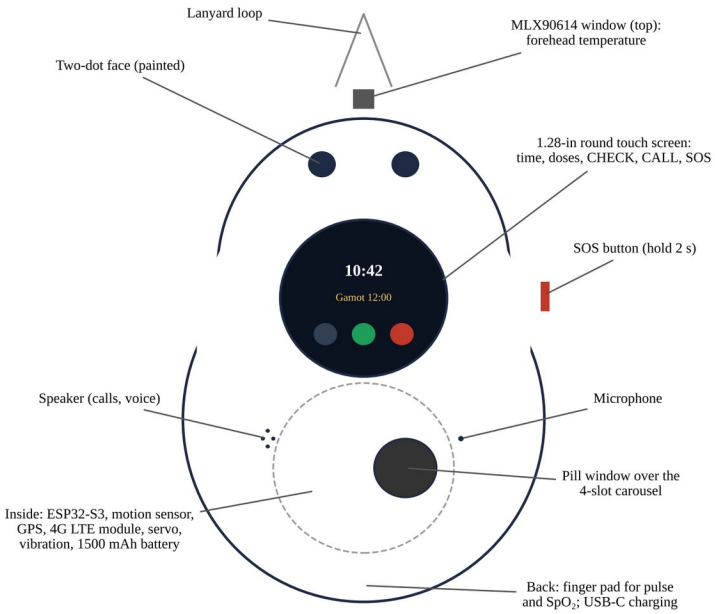
ENGINEERING NOTEBOOK

KALINGA

Development of a 3D-Printed, Baymax-Inspired Touch-Screen Health Pendant with GPS and 4G Calling



Front view, actual size about 66 × 91 × 43 mm, 3D-printed in white PLA.

The pendant to be built (front view).

Team: _____, _____, _____ Adviser: _____ School: _____

How to keep this notebook. Write every entry on the day the work is done. Sign and date each entry and have a witness sign. Do not erase or tear pages: cross out mistakes with one line. Paste printouts, photos, and screenshots with the date. Record what failed as carefully as what worked; judges look for the design process, not only the final product.

Entry 1: Problem and Design Criteria

Date: _____ Recorded by: _____ Witnessed by: _____

Problem. Older adults and people with chronic illness are often alone at home while their families work. Falls go unnoticed, doses are missed, some older adults get lost, and fevers rise unseen. Existing products each do one job and cost more together.

Goal. Build one low-cost touch-screen wearable that checks health signs, presents each dose on time, shows location, calls the family, and shows everything in a phone app.

Criterion	Target	How it will be checked
Cost	Below PHP 5,000 for parts	Receipts (Appendix A.3)
Size and weight	Fits in a hand; about 66 × 91 × 43 mm; under 160 g	Ruler and kitchen scale

GPS	Error under 10 m under open sky	Surveyed points
Calls	At least 95% of calls connect; clarity 4 of 5	40 test calls
Touch screen	Every action in 3 taps or fewer	Usability session
Temperature accuracy	Bias within ±0.3 °C after calibration	Bland–Altman vs clinic thermometer
SpO_2 accuracy	Arms ≤ 4% (ISO 80601-2-61)	Bland–Altman vs reference oximeter
Fall detection	Sensitivity and specificity ≥ 90%	Mat falls and daily activities
Alert time	Under 60 s to the app	Stopwatch and app time stamps
Doses	Correct slot every time; within 1 min of schedule	5-day dose log
Battery	At least 12 h per charge with GPS and 4G	Battery log
Ease of use	SUS above 68	Questionnaire

Entry 2: Background Research

Date: _____ Recorded by: _____ Witnessed by: _____

- Pulse oximetry: red and infrared light through the fingertip give SpO_2 and pulse rate (Allen, 2007; Jubran, 2015). Readings drop in accuracy with movement, cold fingers, nail polish, and darker skin.
- Infrared thermometry: the MLX90614 reads skin temperature without contact; forehead skin is cooler than the body core, so a calibration equation is needed (Fraden, 2016).
- Falls: free fall (< 0.5 g), then impact (> 2.5 g), then stillness is a simple, proven rule (Bourke et al., 2007).
- Medicine: about half of long-term patients do not take medicines as prescribed; timed reminders help (WHO, 2003).
- Data: the Firebase Realtime Database has a REST interface that a microcontroller can call over HTTPS and a web app can stream live (Google, n.d.). Health data must be protected (Republic Act No. 10173, 2012).

Sources read by the team and notes: _____

Entry 3: Ideas and Decision Matrix

Date: _____ Recorded by: _____ Witnessed by: _____

Each idea is scored 1 (poor) to 5 (best) on each criterion and multiplied by the weight. The scores below are the starting estimates; change them after your own tests and sign the decision.

Option for each function	Cost ×3	Size ×2	Ease to build ×2	Reliability ×3	Total
Dispensing: 4-slot carousel + SG90 servo	5	3	4	4	41
Dispensing: single lid with servo latch	5	4	4	3	40
Dispensing: sliding drawer + linear servo	3	4	2	3	30
Temperature: MLX90614 non-contact	3	4	5	4	39
Temperature: MAX30205 skin contact	4	5	3	3	37
Link: Wi-Fi + Firebase + web app	5	5	3	4	43
Link: Bluetooth to the older adult's phone	5	5	3	2	37
Link: GSM SMS module	3	2	3	3	28
Wi-Fi app + 4G calls and SMS (A7670)	3	3	3	5	37
Screen: 1.28-in round touch board (all-in-one)	3	5	4	4	39
Screen: 0.91-in OLED + buttons	5	4	4	3	40

Chosen design: carousel + SG90, MLX90614, round touch board, Wi-Fi + Firebase app with 4G calls and SMS (the touch board was chosen over the OLED because older adults need large buttons). Team signatures: _____

Entry 4: Bill of Materials and Tools

Date: _____ Recorded by: _____ Witnessed by: _____

Item	Qty	Est. PHP
Waveshare ESP32-S3 1.28-in round touch LCD board	1	1,300
MLX90614 (GY-906) module	1	550
MAX30102 module	1	180
ATGM336H GPS module with patch antenna	1	300
SIMCom A7670 4G LTE board (VoLTE) with antenna	1	1,200
Prepaid nano-SIM	1	50
PAM8302 amplifier, 8 Ω 1 W speaker, electret microphone	1 set	200
SG90 micro servo	1	120
LiPo 3.7 V 1500 mAh	1	280
Coin vibration motor, AO3400, S8050, 1000 µF, resistors	1 set	80
SOS button, slide switch, perfboard, wires	1 lot	110
PLA filament or print service (about 75 g)	1 lot	300
Black acrylic paint, sandpaper	1 lot	40
Breakaway lanyard, M2 screws	1 set	60
TOTAL (estimated)		**4,770**

Tools. 3D printer (FDM, 0.4-mm nozzle) or a print shop; slicer (UltiMaker Cura, PrusaSlicer, or Bambu Studio); soldering iron, solder, flux; wire stripper; multimeter; hot-glue gun; small files and sandpaper (220 and 400); flush cutters; laptop with Arduino IDE 2; USB-C cable; placebo candy tablets for testing.

Entry 5: CAD Design of the Shell

Date: _____ Recorded by: _____ Witnessed by: _____

The shell is an egg-shaped body (two spheres joined and flattened) split into a front and a back shell, plus a 36-mm pill carousel. The model is a Python script, so every size can be changed and the STL files regenerated:

pip install manifold3d trimesh then *python cad/kalinga_pendant.py*

Feature	Size (mm)	Why
Wall thickness	2.0	Strong enough for PLA; light
Overall	60 × 83 × 39	Fits the SG90 servo and carousel
OLED window	24 × 8	0.91-in OLED active area 22.4 × 5.6
Finger pad window	13 × 13	MAX30102 module
Thermometer window (top)	Ø 9.8	MLX90614 TO-39 can, Ø 9.2
Pill window	Ø 11	Pocket Ø 9.6 under it
Carousel	Ø 36 × 7, 4 pockets	Pockets at 45°, 90°, 135°, 180°; solid at 0° = closed
Button holes	Ø 6.6	6 × 6 mm tactile button caps
USB-C and switch slots	10 × 4, 9 × 4	TP4056 port, slide switch
Lip	3 deep, 1.2 thick	Locates the front shell on the back shell

Front shell (face)



Front shell rendered from front_shell.stl.

Exploded view: back shell, carousel, front shell



Exploded view: back shell with servo cradle, pill carousel, and front shell.

Changes made to the CAD and why: _____

Entry 6: 3D Printing

Date: _____ Recorded by: _____ Witnessed by: _____

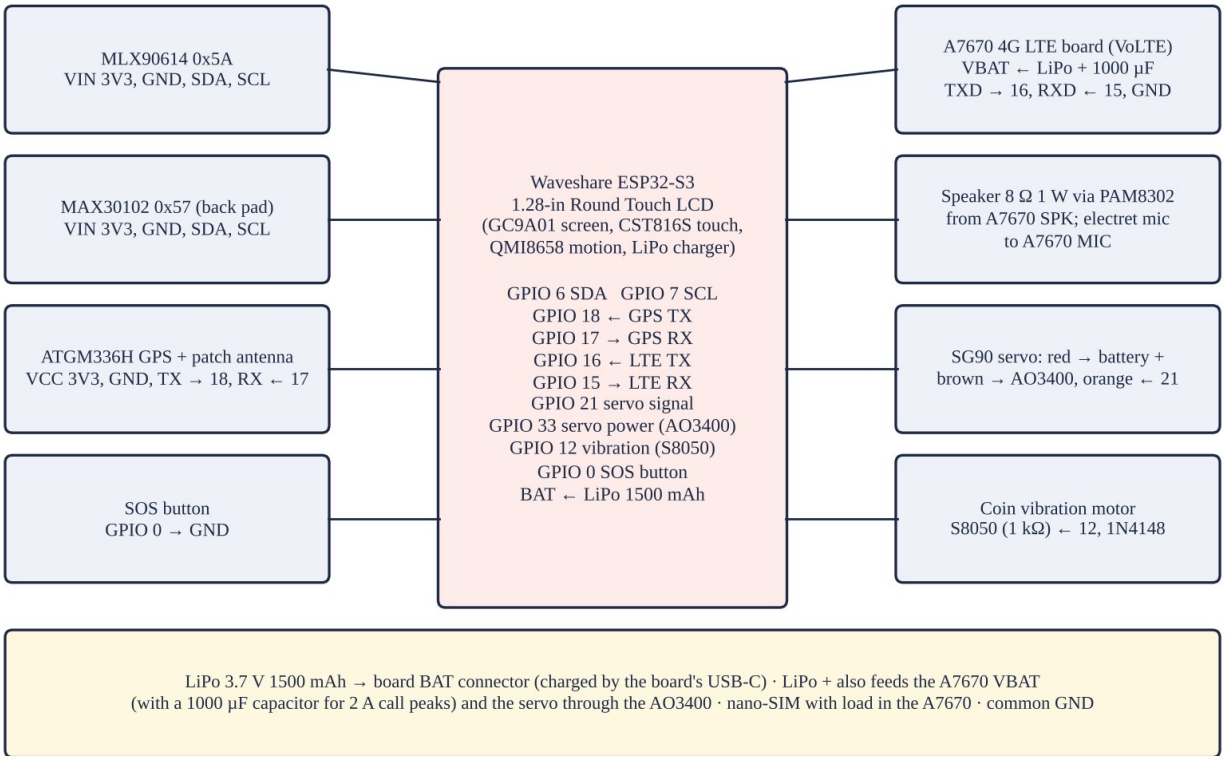
Setting	Shells	Carousel
Material	White PLA	White or red PLA
Nozzle / layer height	0.4 mm / 0.2 mm	0.4 mm / 0.2 mm
Walls / top and bottom layers	3 / 4	3 / 4
Infill	20% gyroid	30% gyroid
Nozzle / bed temperature	205 °C / 60 °C	205 °C / 60 °C
Orientation	Flat split edge on the bed (open side down)	Flat, pockets facing up
Supports	Tree supports, everywhere; 5-mm brim	None
Estimated time and filament (from slicer)	_____ h, _____ g	_____ h, _____ g

1. Print the carousel first as a test: check that a placebo tablet fits each pocket and that the SG90 horn sits flat in its recess.
2. Print the back shell, then the front shell. Watch the first layer; stop and level the bed if lines do not stick.
3. Remove supports with flush cutters and pliers. Sand edges with 220 then 400 grit.
4. Dry-fit the two shells: the lip must slide in with light pressure. If tight, sand the lip; if loose, add one layer of tape.
5. Paint the two eye recesses with black acrylic; let dry for 1 hour. Keep the face design original (two dots, no copied logos) so it can be filed as an Industrial Design.
6. Weigh and measure each part and paste photos here.

Part	Print date	Time (h)	Mass (g)	Problems seen	Fix

Entry 7: Electronics Assembly

Date: _____ Recorded by: _____ Witnessed by: _____



Wiring diagram.

Part	Connects to (ESP32-C3)
MLX90614, MAX30102 (and on-board touch, QMI8658)	GPIO 6 / 7 — I²C at 100 kHz
ATGM336H GPS	GPIO 18 / 17 — UART1, 9600 baud
A7670 4G LTE	GPIO 16 / 15 — UART2, 115200 baud; VBAT from LiPo + 1000 µF
SG90 servo	GPIO 21 / 33 — AO3400 switches the servo ground
Vibration motor	GPIO 12 — 1 kΩ; 1N4148 across motor
SOS button	GPIO 0 — Also the BOOT button
LiPo 1500 mAh	BAT connector — Charged by the board's USB-C

1. Check the touch board alone first: upload a GC9A01 example from the GFX Library for Arduino and a CST816S touch example. Then run an I²C scan: expect 0x15 (touch), 0x6B (QMI8658), 0x5A (MLX90614), and 0x57 (MAX30102).
2. GPS: connect TX → GPIO 18 and RX ← GPIO 17; outdoors, the TinyGPSPlus example must show 6 or more satellites.
3. 4G: insert the nano-SIM, connect TXD → GPIO 16 and RXD ← GPIO 15, VBAT to the LiPo with a 1000 µF capacitor. Send AT, AT+CSQ (signal), and ATD+639…; to test a call. Wire the speaker through the PAM8302 and the microphone.

4. Build the small parts on perfboard: S8050 with 1 kΩ base resistor and 1N4148 across the vibration motor; AO3400 with 10 kΩ gate-to-ground resistor for the servo.
5. LiPo safety: connect the battery to the TP4056 B+ and B– only, with the correct polarity, and never short, bend, or puncture it. Charge only under supervision.
6. Plug the LiPo into the board's BAT connector through the slide switch; the board charges it by USB-C.
7. Measure with the multimeter before powering: battery 3.7–4.2 V; no short between 5V and GND.
8. Record each test below.

Module	Test	Expected	Result	OK?

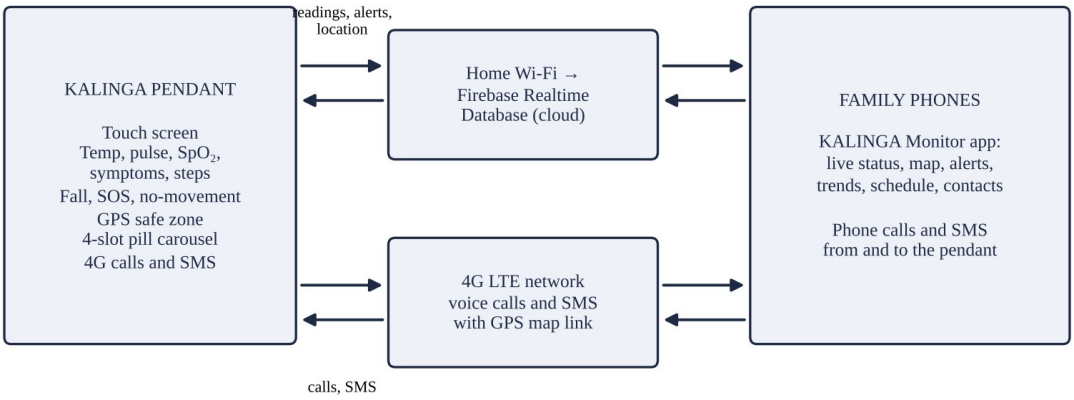
Entry 8: Programming the Pendant

Date: _____ Recorded by: _____ Witnessed by: _____

1. Install Arduino IDE 2 and “esp32 by Espressif Systems” in Boards Manager.
2. Install libraries: GFX Library for Arduino, CST816S, TinyGPSPlus, Adafruit MLX90614, SparkFun MAX3010x, ESP32Servo, ArduinoJson.
3. Open firmware/kalinga_pendant/kalinga_pendant.ino. Type the Wi-Fi name and password, the Firebase Web API key and database URL, the pendant's account email and password, and the patient id.
4. Board: ESP32S3 Dev Module; USB CDC On Boot: Enabled; Flash Size and PSRAM as on the Waveshare wiki. Upload.
5. Check: the round screen shows the face, time, and CHECK, CALL, SOS buttons; the carousel closes; holding SOS for 2 s sends an app alert and an SMS with a map link.

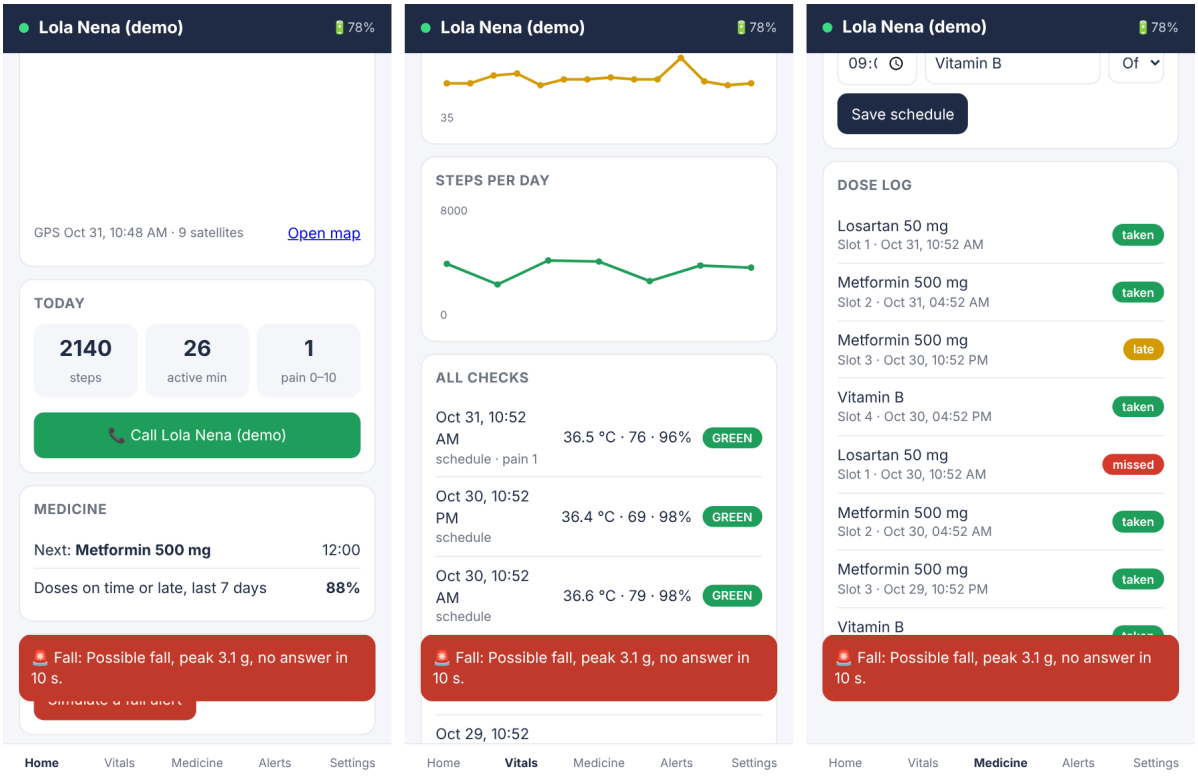
Entry 9: Firebase and the KALINGA Monitor App

Date: _____ Recorded by: _____ Witnessed by: _____



How the pendant, the database, and the app connect.

1. Go to console.firebase.google.com, create a project, and add a Web app. Copy the Web API key.
2. Authentication → Sign-in method → enable Email/Password. Add one user for the pendant and one for each family member. Copy each user's UID.
3. Realtime Database → Create (location: Singapore, asia-southeast1) → start in locked mode. Copy the database URL.
4. Rules: paste app/database.rules.json and publish.
5. Data: add members → <patient id> → <each UID> = true (for the pendant and every family member).
6. Host the app folder on GitHub Pages or Firebase Hosting (HTTPS is needed to install it). Open it on a phone, add it to the home screen, enter the key, URL, patient id, email, and password, and sign in.
7. In the app's Settings, enter the pendant's SIM number, up to three family contacts, and the safe-zone radius; when the pendant has a GPS fix, tap Set home.
8. Use “Try the demo” to show the app to judges without the pendant.



KALINGA Monitor screens (demo mode).

Entry 10: Final Assembly

Date: _____ Recorded by: _____ Witnessed by: _____

1. Set the servo to the closed position (power on once; the program moves it to 0°) before fitting the horn and carousel, so the solid sector sits under the window.
2. Fix the servo in the back-shell cradle with a drop of hot glue; screw the single-arm horn into the carousel recess.
3. Fix the round touch board behind the 33-mm window with foam tape, the MAX30102 behind the back finger window, and the MLX90614 can in the top hole.
4. Place the GPS patch antenna near the top, facing up; stick the 4G antenna along the inside of the back shell; fit the speaker behind the grille and the microphone behind its hole; add the battery with foam tape.
5. Fit the buttons and caps, close the shells, and add two M2 screws or a drop of glue at the bottom.
6. Thread the breakaway lanyard. Fill the four pockets with placebo tablets and run one dose cycle.

Entry 11: Calibration

Date: _____ Recorded by: _____ Witnessed by: _____

1. Bench test of the thermometer with a black-painted can in a water bath at 35.0, 36.0, 37.0, 37.5, 38.0, and 40.0 °C against a probe thermometer.
2. Calibration set (10 volunteers, after IRB approval): TEMP_A = SLOPE(reference, raw), TEMP_B = INTERCEPT(reference, raw); PR and SpO_2 offsets = mean of (reference − pendant).
3. Fall thresholds: 10 mat falls and 10 daily activities by the researchers; adjust FREE_FALL_G and IMPACT_G, then keep them fixed.

Target or volunteer	Reference	Pendant raw	Pendant calibrated	Difference

Entry 12: Test Records

Date: _____ Recorded by: _____ Witnessed by: _____

Falls and daily activities

Trial	Event	Detected?	Cancelled?	Peak g	Alert time (s)

Doses

Date	Slot	Scheduled	Opened at	Recorded as	Observer says

GPS and calls

Point or call	Reference	Pendant result	Error (m) or connected	Clarity 1–5

Battery

Charge no.	Full at	Empty at	Hours	Notes

Entry 13: Problems and Iterations

Date: _____ Recorded by: _____ Witnessed by: _____

Write each problem the day it happens, what caused it, what you changed, and what happened after.

Date	Problem	Cause	Change made	Result

Entry 14: Safety Checklist and Reflection

Date: _____ Recorded by: _____ Witnessed by: _____

- Adult supervision for the 3D printer, soldering, and LiPo charging.
- IRB approval and signed consent before any test with people; placebo tablets only.
- Falls simulated only by the researchers on 20-cm mats with pads.
- Breakaway lanyard so the cord cannot strangle the wearer.
- Firebase locked by rules; coded patient id; data deleted after the fair.

What worked: _____

What we would change next: _____